



Haoyu Ren, PhD Candidate

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EDUCATION

University of Maryland, Baltimore County, Baltimore, MD

M.S. in Mathematics and Statistics, 2024

Ph.D. in Mathematics and Statistics, GPA: 3.8/4.0 | September 2021 – Present

Expected graduation: 2026

Advisors: Dr. Yi Huang and Dr. Chixiang Chen

University of Maryland, College Park, College Park, MD

B.S. in Mathematics and Statistics, cum laude, GPA: 3.9/4.0

EXPERIENCE

Research Assistant | June 2024 – Present

Division of Biostatistics and Bioinformatics, Department of Epidemiology and Public Health, University of Maryland, School of Medicine

Collaborating with University of Maryland, School of Medicine

- Collaborated with clinicians and cross-functional research teams to translate clinical questions into appropriate statistical models, conduct data analyses, and interpret study findings.

Causal Machine Learning

- Co-led the development of a new doubly robust method for continuous treatments while addressing missing data using a change-point model.
- Co-led the development of a new method for analyzing heterogeneous treatment effects under continuous treatment while addressing missing data using a varying-coefficient model.

Causal Meta-Analysis

- Co-led the development of a new weighting function with smaller bias and variance under rare events than existing weighting functions.

Teaching Assistant | September 2021 – May 2024

Department of Statistics, University of Maryland, Baltimore County

- Explained mathematical and statistical concepts to students.
- Held office hours and created and graded quizzes and homework assignments.

Biostatistics Intern | June 2026 – August 2026

Biostatistics and Research Decision Sciences, Merck, Upper Gwynedd, PA

- Evaluated the impact of imaging frequency on progression-free survival (PFS) estimation and hypothesis testing in oncology trials.

PUBLICATIONS

- Ren, H., Wen, T., ..., Chen, C. *A Doubly Robust Approach to Evaluating the Impact of Hospital Length of Stay on Post-Fracture Recovery in Older Adults with Cognitive Impairment*. Under revision at *Statistics in Medicine*.
- Ren, H., ..., Chen, C. *A Novel Causal Machine Learning Approach to Study Threshold Effects of Continuous Exposure Using Multi-Source Linked Medicare Claims Data*. Prepared for submission to *Biostatistics*.
- Shen, B., Ren, H., Shardell, M., Falvey, J., & Chen, C. (2023). *Analyzing risk factors for post-acute recovery in older adults with Alzheimer's disease and related dementia: A new semiparametric model for large-scale Medicare claims*. *Statistics in Medicine*, 43(5), 1003–1018. <https://doi.org/10.1002/sim.9982>
Contribution: Coded and evaluated the method through simulation.
- Falvey, J. R., Chen, C., ..., Ren, H., ... (2023). *Associations of days spent at home before hip fracture with postfracture days at home and 1-year mortality among Medicare beneficiaries living with Alzheimer's disease or related dementias*. *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences*, 78(12), 2356–2362. <https://doi.org/10.1093/gerona/glad158>
Contribution: Applied clustering methods to analyze multiple longitudinal outcomes.
- Kardon, A., Kim, D., Ren, H., ... (2024). *A propensity-score weighted analysis of short-term corticosteroid therapy for refractory pain following spontaneous subarachnoid hemorrhage*. *Neurocritical Care*.
Contribution: Applied propensity-score weighting to compare treatment effects.

- Graeme, F., Chen, C., ..., **Ren, H.**, ... (2025). *Microbubble-enhanced focused ultrasound with temozolomide for patients with high-grade gliomas: A multicenter, open-label, phase 1/2 trial*. Lancet Oncology.
Contribution: Analyzed cfDNA and fragment length ratio data.
- Kardon, A., **Ren, H.**, ... (2025). *Early markers of systemic inflammation are not related to pain burden after aneurysmal subarachnoid hemorrhage: A multicenter observational study*. Neurocritical Care. DOI: 10.1007/s12028-025-02396-w.
Contribution: Conducted variable selection and tested the proportional odds assumption for ordinal logistic regression.
- Fahim, A., ..., **Ren, H.**, ... (2025). *Systemic Cyst(e)inase Administration Induces Ferroptosis and Synergizes with Temozolomide in Glioblastoma*. iScience. <https://doi.org/10.1016/j.isci.2025.114350>
- Blair, R., ..., **Ren, H.**, ... (2026). *Post-transcriptional control of HIF-1 α by MBNL1 restrains hypoxia-driven stemness in GBM*. Neoplasia. <https://doi.org/10.1016/j.neo.2025.101262>
Contribution: Applied a linear mixed-effects model to analyze the performance of three genes across different cell types and conditions.

MANUSCRIPTS IN PREPARATION

- Harrison, D. M., ..., **Ren, H.**, Chen, C., ... (2026). *Cellular-level Quantification of the Impact of Multiple Sclerosis on Retinal Ganglion Cell Bodies and Axons Using Adaptive Optics–Optical Coherence Tomography*. Manuscript in preparation.
Contribution: Compared retinal cell and RNFL densities across MS subgroups and evaluated inter-rater reliability and clinical associations.
- Falvey, J., ..., **Ren, H.**, ... (2026). *Engaging Community and Municipal Stakeholders to Improve High Quality Aging in Place after Hip Fracture*. Manuscript in preparation.
Contribution: Performed XGBoost variable selection, model comparison, and odds-ratio analysis.
- Morris, N., ..., **Ren, H.**, ... (2026). *The Relationship Between Blood and Edema Clearance with Pain Burden after Aneurysmal Subarachnoid Hemorrhage: A Retrospective Cohort Study*. Manuscript in preparation.
Contribution: Evaluated associations between CT hemorrhage-score changes and pain outcomes using adjusted and sensitivity analyses with G-formula.
- Wen, T., ..., **Ren, H.**, ... (2026). *Hospital Characteristics and Recovery after Hip Fracture in the United States*. Manuscript in preparation.
Contribution: Developed a risk-adjusted XGBoost/SHAP ensemble to rank hospital characteristics and predict days at home.

AWARDS

- Dean's List, Fall 2020 and Spring 2020, University of Maryland, College Park
- Graduated cum laude, University of Maryland, College Park

PRESENTATIONS

- ENAR, New Orleans, March 2025.
Contributed Paper: Breaking Barriers in Statistical Insights for Health Disparities and Outcomes
A User-Friendly Approach to Detecting the Heterogeneity of Hospital Length of Stay Effects on Post-Acute Recovery Outcomes for Older Adults with Cognitive Conditions
- JSM, Boston, August 2026.
Contributed Paper: Section on Statistics in Epidemiology — Causal Inference Methods and Applications in Epidemiology
A Novel Causal Machine Learning Approach to Study Threshold Effects of Continuous Exposure
- ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Maryland, 2026.
Poster
A Novel Causal Machine Learning Approach to Study Threshold Effects of Continuous Exposure

SKILLS

- **Programming and Software:** MATLAB, SAS, R, Java, C++, Minitab
- **Other:** Word, Excel, PowerPoint, cross-functional collaboration, statistical modeling, and study design